Routing Visualization and Surface Water Budgeting Tool for Groundwater-Surface Water Models

Matthew Baillie, PG, CHg and Lauren Sather, PhD, West Yost Associates

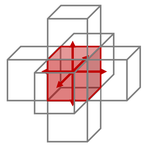
Introduction

Integrated groundwater-surface water models are important tools for understanding basin-scale hydrologic dynamics. In groundwater-focused tools like MODFLOW surface water routing is handled differently from groundwater flow, which can occur only between neighboring cells. In contrast, surface water routing is defined by the modeler and can connect non-adjacent locations (e.g., to represent pipeline transfers). This leads to two challenges: 1) it is easy to make routing errors, and 2) traditional zonal groundwater budget tools (e.g., ZoneBudget) cannot be used to create zonal surface water budgets.

To address these limitations, we developed two Python tools. The first generates a stream network visualization with all routing connections shown. This allows the user to identify routing issues, such as isolated streams, hanging streams, improper segment sequencing, and misrouted connections. The second computes a zonal surface water budget. For MODFLOW applications, zone interconnections are automatically identified and SFR and FMP diversions are tracked separately.

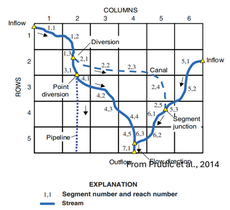
Groundwater Flow and ZB

Within a gridded representation of a groundwater system, flow can only be exchanged between adjacent cells. In a rectilinear finite difference grid (such as used by classic MODFLOW), a cell can exchange flow with up to six adjacent cells; the user does not specify these connections. Exchanges are written to the cell-by-cell (cbc) flow file. Developing zonal groundwater budgets from this output is straightforward; ZoneBudget (Harbaugh, 1990) requires a file of zone numbers for each model cell and the cbc file to generate a water budget, and connections between zones are automatically determined.



Streamflow Routing

Reach-based streamflow routing, such as is used in the Streamflow-Routing Package (SFR; Prudic et al., 2004), relies on the user to define how the stream network is interconnected and how water moves through it. Interconnections are not necessarily obvious based on streams' physical locations, as water can move between non-adjacent streams in some systems (e.g., via pipeline transfers; see figure to right). Available visualization approaches for MODFLOW (including GUIs) show the physical locations of streams, not how they are connected, and do not provide the user sufficient detail to QC routing.

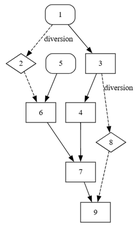


From Prudic et al., 2014

An SFR user has significant freedom in defining streamflow routing, and can create routing patterns that are physically unreasonable without generating errors or warnings.

Routing Visualization

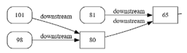
We wrote a Python script that reads a MODFLOW SFR input file and generates a directed graph of the streamflow routing network using the dot language (Gansner and North, 1999). This visualization does not use spatial information (e.g., stream segment locations) to generate the network; it is not a realistic map of the stream network. The figure to the right shows one possible visualization for the stream network depicted in the map above.



This type of visualization can aid QC of stream network routing where other visualization approaches fall short. This type of stream network visualization can quickly help a model developer (or model reviewer) identify issues with the stream routing.

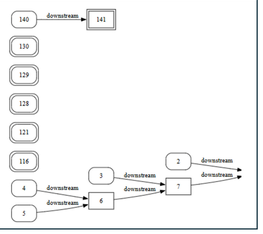
Improper Segment Sequencing

SFR routes flow through the stream network sequentially from the first segment to the highest number segment. Assigning outflow from a higher-numbered segment to a lower-numbered segment in MODFLOW will not cause an error.* To the right is an example of improper sequencing from a model.



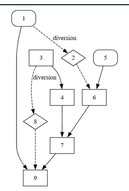
Hanging and Isolated Streams

If a SFR segment does not have a downstream segment defined, its outflow is removed from the model. This is proper for segments that represent locations where streams flow out of the model. Hanging streams (which do not have an outlet channel) are also commonly defined this way. Having hanging or isolated streams could result in water balance issues in the model, which would not be obvious from analysis of groundwater head maps. To the right is an example of hanging streams in a model.



Misrouted Connections

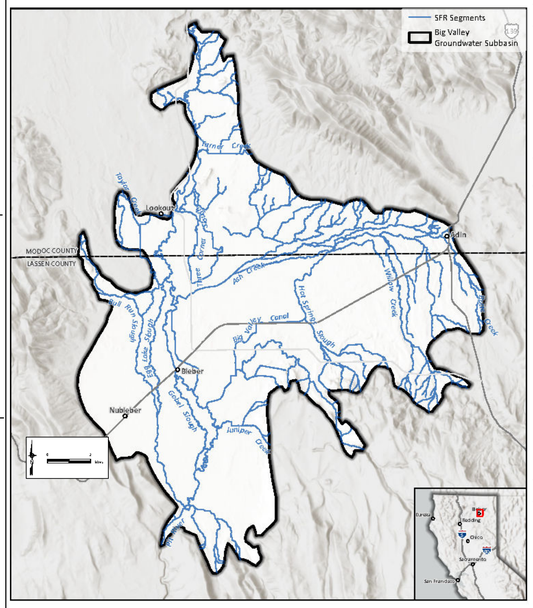
Network routing is defined within the SFR file, which could contain routing errors resulting from user error, including simple typos. To the right is an example of a misrouted connection (Segment 1 misrouted to Segment 9 instead of 3).



* Prudic et al. (2004) state: "*As long as the upstream segments have a lower number than the downstream segment, outflows from the upstream segments will be added correctly as inflow to the downstream segment. If tributary segments have higher numbers than the connecting downstream segments, then outflows from the higher numbered tributary segments will be from the previous iteration during the MODFLOW solution of ground-water heads for each time step. As long as the difference in head between iterations is small (a small specified closure criteria), the error caused by adding outflow from the previous iteration in the higher numbered tributary segments to the lower numbered connecting segments will be small.*"

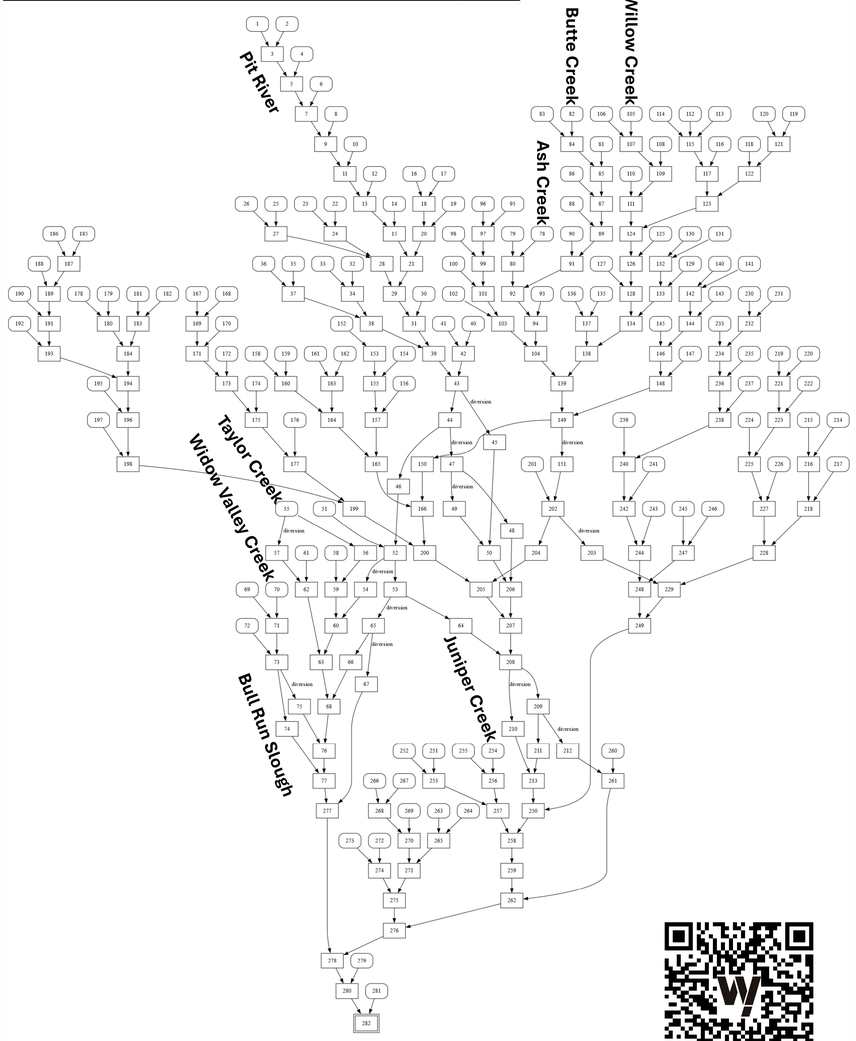
Routing Example: Big Valley Basin, CA

The Big Valley Basin in northeastern California is a medium-priority groundwater basin under SGMA. Basin agriculture relies on both groundwater and surface water, which is conveyed via canals and sloughs from sources (the Pit River and its tributaries) to points of use. Groundwater is used near where it is pumped. Fields may rely on groundwater, surface water, both, or neither for irrigation supply. Stream routing is critical to understanding water use and availability in the basin.



West Yost prepared a groundwater-surface water interaction model of the basin in MODFLOW-OWHMv2 (Boyce et al., 2020) to assist with SGMA compliance. The Big Valley Basin Hydrologic Model (BVBHM) represents major features stream network features (figure at left), including diversions to conveyance structures (simulated using SFR) serving multiple points of diversion (PODs), and individual PODs (simulated using the Farm Process, FMP; Boyce et al., 2020).

The routing network is depicted below. This shows not only how the streamflow network is connected, but the direction of flow through conveyance structures.



Access the scripts at https://github.com/mnbaillie/Scripts or via QR code

SFR ZoneBudget

MODFLOW has lacked a zonal stream network water budget capability equivalent to ZoneBudget because:

- Stream network routing connections are more complicated than between finite difference cells
- Surface water can be simulated using several different packages, including RIV, STR, SFR, SWR, and LAK
- There was not, until MODFLOW-OWHMv2, an option for surface water output equivalent to the cbc file

We created a Python script that takes as input:

1. The SFR input file (for characterizing the routing)
2. A user-supplied zone file (comma-delimited list of segments or reaches and zone numbers)
3. The output file for SFR (binary or ascii) containing the reach-by-reach streamflow

The script produces a zonal surface water budget for every model timestep, with a tab for each user-defined zone. The script tracks:

- Flow entering the zone at the heads of streams (either at the edge of the model or internal to the zone)
- Flow leaving the model from the zone (no outflow segment defined)
- Runoff and precipitation entering the segment and direct evapotranspiration leaving the segment
- Diversions from the stream network via FMP
- Groundwater-surface water exchange
- SFR diversions internal to the zone (not part of the water budget, but included for QC)
- Exchanges with other zones (in and out components are tracked separately)
- Mass balance error based on the water budget component (included for QC)

Areas with no zone defined are tracked as Zone 0. The user can define zones by segment or by reach. The

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	DATE_TIME	PER	STP	DELT	SIMTIME	ZONE	FLOW_HEAD	FLOW_SEEPAGE	FLOW_OUTOFMODEL	RUNOFF	FARM_DIVERSION_NET_QC	DIVERSION_INTERNAL_QC	PRECIP	STREAM_ET	IN_FROM_ZONE_0	OUT_TO_ZONE_0	IN_FROM_ZONE_1	OUT_TO_ZONE_1
2	1980-10-01 0:00:00	1	1	31	31	5	3.233064191	-116.5042761	0	36.3276797	0	0	0	0	0	0	0	0
3	1980-11-01 0:00:00	2	1	30	61	5	0	-134.960089	0	9.245482433	0	0	0	0	0	0	0	0
4	1980-12-01 0:00:00	3	1	31	92	5	0	-211.0551931	0	36.9317341	0	0	0	0	0	0	0	0
5	1981-01-01 0:00:00	4	1	31	123	5	0	-236.9475997	0	56.63650298	0	0	0	0	0	0	0	0
6	1981-02-01 0:00:00	5	1	28	151	5	0	-276.1017299	0	57.0370327	0	0	0	0	0	0	0	0
7	1981-03-01 0:00:00	6	1	31	182	5	0	-235.102181	0	29.68471762	0	0	0	0	0	0	0	0
8	1981-04-01 0:00:00	7	1	30	212	5	53.23123829	371.8591286	0	14.61234126	66.83191341	119.0082645	0	0	0	0	1785.123967	0
9	1981-05-01 0:00:00	8	1	31	243	5	69.89703736	504.2016251	0	39.81416422	122.6342496	122.9752066	0	0	0	0	1844.628099	0
10	1981-06-01 0:00:00	9	1	30	273	5	50.96628258	683.2130815	0	33.29424866	424.8132945	119.0082645	0	0	0	0	1785.123967	0
11	1981-07-01 0:00:00	10	1	31	304	5	7.927990892	651.4212299	0	37.56141594	430.3982948	122.9752066	0	0	0	0	1696.051506	0
12	1981-08-01 0:00:00	11	1	31	335	5	4.793470292	114.1410104	0	51.51940619	351.3493536	56.03384576	0	0	0	0	412.3172695	0
13	1981-09-01 0:00:00	12	1	30	365	5	7.649429883	252.4778892	0	38.72908271	97.40396278	119.0082645	0	0	0	0	1378.96348	0
14	1981-10-01 0:00:00	13	1	31	396	5	3.233064191	-322.6368554	0	311.1169075	0	0	0	0	0	0	0	0
15	1981-11-01 0:00:00	14	1	30	426	5	0	-326.9035883	0	108.595931	0	0	0	0	0	0	0	0
16	1981-12-01 0:00:00	15	1	31	457	5	0	-283.6706995	0	25.48339941	0	0	0	0	0	0	0	0

This output not only quantifies water moving through the stream network, but also provides valuable QC insight. The modeler can use this output to assess:

- Are the connections between zones as expected? Is the direction of inter-zone flow as expected?
- Is the amount of streamflow leaving the model appropriate? Do the outputs indicate that streamflow disappearance from the ends of hanging streams is problematic?
- Are diversions taking the expected amount of streamflow?

This zonal water budget supplements existing tools (e.g., the Gage package) to provide more complete insight to the behavior of the stream network in a MODFLOW model.

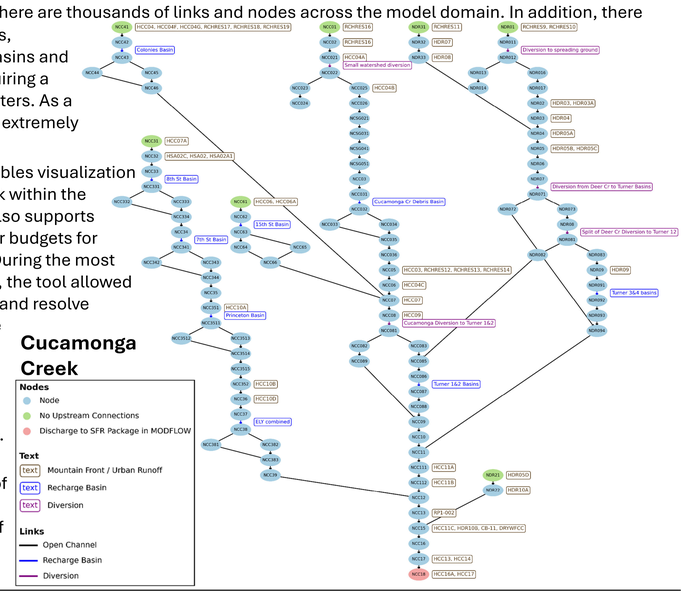
R4 Example

The diagram to the right shows the link/node network for Cucamonga Creek produced using the visualization tool. Cucamonga Creek is one of seven creek systems represented in the rainfall–runoff–routing–recharge (R4) model developed for the Chino Basin.

The R4 model relies on link and node text files, where users define each link and node along with their specific properties. There are thousands of links and nodes across the model domain. In addition, there are multiple link types, including recharge basins and diversions, each requiring a unique set of parameters. As a result, these files are extremely prone to errors.

This tool not only enables visualization of the stream network within the model domain, but also supports development of water budgets for each creek system. During the most recent model update, the tool allowed us to quickly identify and resolve multiple issues in the link and node files, including hanging nodes, incorrect linkages, and mis-classification of links.

This application shows the flexibility of these scripts for use with different types of modeling codes.



References

Boyce, S.E., R.T. Hanson, I. Ferguson, W. Schmid, W. Henson, T. Reimann, S.M. Mehl, and M.M. Earll, 2020, One-Water Hydrologic Flow Model: A MODFLOW Based Conjunctive-Use Simulation Software. *U.S. Geological Survey Techniques and Methods 6-A60*. 458p.

Gansner, E.R. and S.C. North, 1999, An Open Graph Visualization System and Its Applications to Software Engineering, Software- Practice and Experience, 00(S1), p.1-5.

Harbaugh, A.W., 1990, A Computer Program for Calculating Subregional Water Budgets Using Results from the U.S. Geological Survey Modular Three-Dimensional Finite-Difference Ground-Water Flow Model, *U.S. Geological Survey Open-File Report 90-392*, 49p.

Prudic, D.E., L.F. Konikow, and E.R. Bant, 2004, A New Streamflow-Routing (SFR1) Package to Simulate Stream-Aquifer Interaction with MODFLOW-2000, *U.S. Geological Survey Open-File Report 2004-1042*, 104p.